

Chapter 6: Graphical Representation of Causal Effects (Causal DAGs)

Applied Use Case: *Did the Retention Offer Actually Reduce Churn?*

1 Introduction to Causal Directed Acyclic Graphs (DAGs)

Reference: *What If (Hernán & Robins), Section 6.1, pp. 69–71*

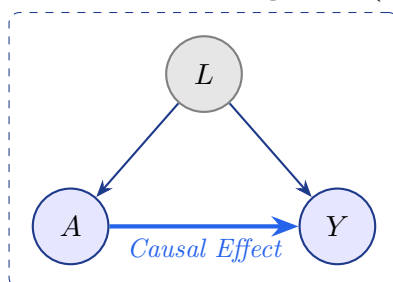
A **Causal Directed Acyclic Graph (DAG)** is a formal mathematical tool used to visually map qualitative causal assumptions between variables in a study population.

- **Nodes (Vertices):** Represent variables (treatments, outcomes, covariates, unmeasured factors).
- **Directed Edges (Arrows):** Represent direct causal relationships from cause to effect ($A \rightarrow Y$).
- **Acyclic:** Time flows strictly forward; no variable can cause itself (no causal feedback loops in the same time slice).

Notation in Our Churn Use Case

- **A: Treatment** – Customer received a retention offer ($A = 1$) vs. no offer ($A = 0$).
- **Y: Outcome** – Customer churned ($Y = 1$) vs. retained ($Y = 0$).
- **L: Measured Baseline Confounder** – Customer complaint history / drop in portal activity.
- **U: Unmeasured Confounder** – Unobserved budget shift / unrecorded competitor discount.
- **M: Mediator** – Post-offer portal activity (number of log-ins).
- **S: Selection / Collider** – Customer completed a post-campaign satisfaction survey ($S = 1$).

Figure 6.1: Basic Confounding DAG ($A \leftarrow L \rightarrow Y$)



2 The Three Fundamental Building Blocks of Causal DAGs

Reference: *What If, Section 6.3, pp. 73–75; Technical Point 6.1, p. 77*

Every path between treatment A and outcome Y on a causal DAG consists of combinations of three basic structural building blocks:

1. **Chain (Mediator / Sequential Path):** $A \rightarrow M \rightarrow Y$

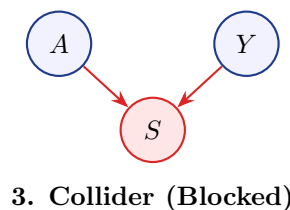
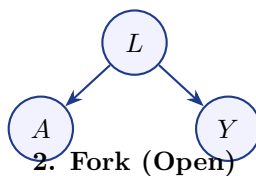
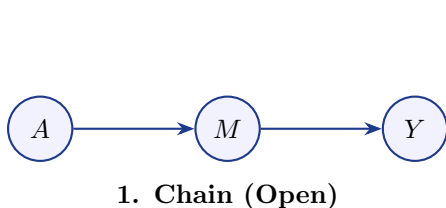
- *Business Interpretation:* The retention offer (A) increases portal log-ins (M), which subsequently reduces churn (Y).
- *Status:* **NATURALLY OPEN**. It carries the true causal effect. Controlling for M **blocks** this indirect path.

2. **Fork (Confounder / Common Cause):** $A \leftarrow L \rightarrow Y$

- *Business Interpretation:* High complaint history (L) leads consultants to issue offers (A), but L also directly increases churn (Y).
- *Status:* **NATURALLY OPEN**. It creates non-causal association (*Backdoor Path*). Controlling for L **blocks** the backdoor path.

3. **Collider (Common Effect / Inverted Fork):** $A \rightarrow S \leftarrow Y$

- *Business Interpretation:* Both receiving an offer (A) and intending to churn (Y) increase the likelihood of filling out a survey (S).
- *Status:* **NATURALLY BLOCKED**. A collider naturally prevents association from flowing between A and Y .
- **CRITICAL WARNING:** Conditioning (filtering/adjusting) on S **OPENS** the path, inducing **Selection Bias**!



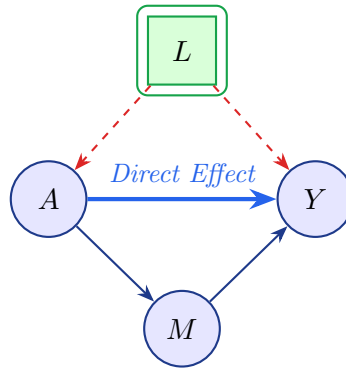
3 The Backdoor Criterion & Confounding Adjustment

Reference: *What If*, Section 6.4, pp. 75–78

To estimate the true population causal effect $E[Y^{a=1} - Y^{a=0}]$, we must identify an adjustment set of variables L that satisfies the **Backdoor Criterion**:

1. L must **block all backdoor paths** between treatment A and outcome Y (paths containing an arrow into A , such as $A \leftarrow L \rightarrow Y$).
2. L must **not contain any descendants** of treatment A (such as mediators M or colliders S).

Adjusted for L (Backdoor Blocked)



Core Rule of Causal Adjustment

Adjusting for L (via IP Weighting or Standardization) successfully blocks the non-causal backdoor path $A \leftarrow L \rightarrow Y$, restoring conditional exchangeability ($Y^a \perp\!\!\!\perp A \mid L$).

4 Unmeasured Confounding & Selection Bias

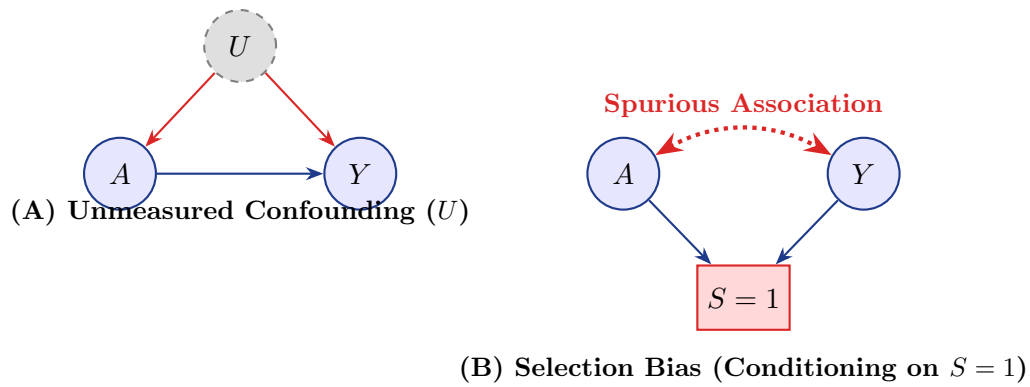
Reference: *What If*, Section 6.5, pp. 78–82; *Fine Point* 6.1

1. Unmeasured Confounding (U)

If there exists an unmeasured variable U (e.g., customer's latent dissatisfaction or external competitor campaign) influencing both offer assignment A and churn Y , the backdoor path $A \leftarrow U \rightarrow Y$ remains **open**. No statistical adjustment on observed data alone can eliminate this bias.

2. Selection Bias / Conditioning on a Collider (S)

Suppose we analyze churn *only among customers who responded to the satisfaction survey* ($S = 1$). Conditioning on S opens the path $A \rightarrow S \leftarrow Y$, introducing spurious correlation between A and Y even if no true causal effect exists!



5 Summary: Structural Rules for Data Scientists

Structural Path	Natural State	Action when Adjusted / Filtered	Methodological Impact
Chain ($A \rightarrow M \rightarrow Y$)	Open	Blocks Causal Path	Misses indirect causal mechanisms
Fork ($A \leftarrow L \rightarrow Y$)	Open	Blocks Backdoor Path	Eliminates Confounding Bias
Collider ($A \rightarrow S \leftarrow Y$)	Blocked	OPENS Path!	Induces Selection Bias